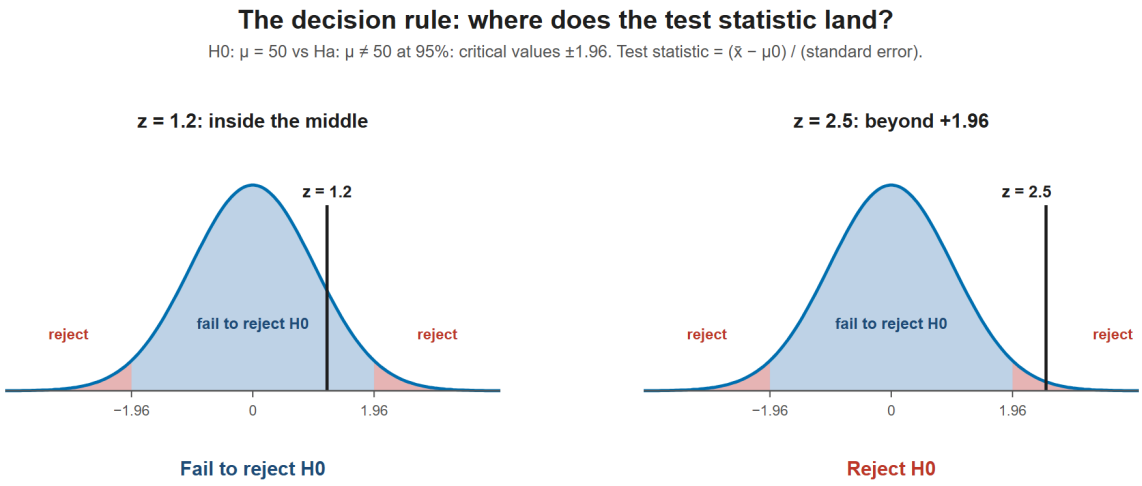
# Lesson 10.4 - Test Statistic for Means and Population Proportions

> **Video:** Test Statistic For Means and Population Proportions by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

## The decision rule

1. Find the **critical value(s)** from the significance level and the type of test. For a two-tailed test at 95%, $\pm 1.96$.
2. Compute the **test statistic** $z_c$ (or $t$) from the sample.
3. Statistic in the rejection region → **reject** $H_0$. Statistic in the fail-to-reject region → **fail to reject** $H_0$.

*Example:* $H_0 : \mu = 50$ vs $H_a : \mu \neq 50$ at 95% has critical values $\pm 1.96$. A calculated $z_c$ beyond 1.96 (or below −1.96) means reject.




Compute the test statistic, then see which region it lands in. Between −1.96 and 1.96, fail to reject H0. Beyond either critical value, reject H0.

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## Test statistic for a mean

| Sample size | $\sigma$ known? | Distribution | Test statistic |
|---|---|---|---|
| $n < 30$ | **No** (only $s$) | **t**, df $= n - 1$ | $t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$ |
| $n < 30$ | Yes | z | $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ |
| $n \geq 30$ | No (use $s$) | z (approximately normal) | $z = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$ |
| $n \geq 30$ | Yes | z | $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ |

Here $\mu$ is the value claimed in $H_0$.

> Many textbooks use the t-distribution **whenever $\sigma$ is unknown**, whatever the sample size. For $n \geq 30$, $t$ and $z$ give nearly the same answer, so the video's rule is a common shortcut.

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## Test statistic for a proportion

$$z = \frac{\hat{p} - p}{\sqrt{\dfrac{pq}{n}}}, \qquad q = 1 - p$$

- $\hat{p}$ = sample proportion ($x/n$)
- $p$ = the proportion claimed in $H_0$ (sometimes written $p_0$). Use this hypothesized value under the square root, not $\hat{p}$.

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## Key takeaways

- Test statistic = (sample value − hypothesized value) ÷ standard error.
- Mean: use $t$ with $s$ when $n < 30$ and $\sigma$ is unknown. Otherwise use $z$.
- Proportion: $z = (\hat{p} - p)/\sqrt{pq/n}$, with $p$ from $H_0$.
- Compare with the critical value(s) to reject or fail to reject $H_0$.